

DOPSoft – Workshop Instructions (HMI Basics)

1. Create a New Project

1. Open **DOPSoft**.
 2. Click **New File**.
 3. Select **HMI Hardware**, then click **Next**.
 4. Select **Communication Port (COM No.)** and **PLC Hardware**.
 5. Click **Finish**.
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Switch (Input Button)

1. Go to **Button** → **Momentary Button**.
 2. Double-click the button.
 3. **Zero State**
 - Select picture: **3D Button – OFF**
 - Enable **Stretch All**
 4. **One State**
 - Select picture: **3D Button – ON**
 - Enable **Stretch All**
 5. Go to **Main** tab.
 6. Enter the **Input Address**.
 7. Click **OK**.
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Output (Indicator Lamp)

1. Go to **Indicator** → **Multistate Indicator**.
 2. Double-click the indicator.
 3. **Zero State**
 - Select picture: **3D Lamp (2-State) – OFF**
 - Enable **Stretch All**
 4. **One State**
 - Select picture: **3D Lamp (2-State) – ON**
 - Enable **Stretch All**
 5. Go to **Main** tab.
 6. Enter the **Output Address**.
 7. Click **OK**.
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Preset Value (Numeric Entry)

1. Go to **Input** → **Numeric Entry**.
 2. Click **OK**.
 3. Enter the **Address**.
 4. If required, set **Gain / Offset / Scaling**.
 5. Click **OK**.
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Accumulator (Numeric Display)

1. Go to **Display** → **Numeric Display**.
 2. Double-click the object.
 3. Enter the **Address**.
 4. If required, set **Gain / Offset / Scaling**.
 5. Click **OK**.
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Text Display (Status Indication)

1. Go to **Indicator** → **Multistate Indicator**.
2. Double-click the object.

State Settings

- **Zero State**
 - Color: **Red** (or any warning color)
- **One State**
 - Color: **Green** (or opposite color for easy understanding)

Text Settings

1. Go to **Text** tab.
2. **Zero State (Language 1)**
 - Text: *Sensor OFF*
3. **One State (Language 1)**
 - Text: *Sensor ON*

Address

1. Go to **Main** tab.
 2. Enter the **Address** (e.g., **X2**, same as shown in previous examples).
 3. Make sure the address matches the actual PLC input/output.
 4. Click **OK**.
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Simulation Tools Instructions (DOPSoft)

Enable Simulation Mode

1. Go to the **Tools** menu.
 2. Select **Simulation**.
 3. Click **On-Line Simulation**.
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Start Simulation

1. Click **Simulation** → **Start**
(or press the *Simulation Start icon on the toolbar*).
 2. The HMI screen will appear in **simulation mode**.
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Using Simulation Tools

1. Test Input Devices (Buttons / Numeric Entry)

- Click the **Momentary Button** on the simulated HMI.
 - Observe the corresponding **PLC input bit** changing.
 - For **Numeric Entry**, enter a value and press **Enter**.
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2. Monitor Output Devices (Indicators / Lamps)

- When the logic condition is true:
 - The **Indicator Lamp** should turn **ON**.
 - When the condition is false:
 - The **Indicator Lamp** should turn **OFF**.
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3. Text Display Verification

- When the address bit = **0**
 - Text shows: **Sensor OFF**
 - Color shows: **Red** (or warning color)
 - When the address bit = **1**
 - Text shows: **Sensor ON**
 - Color changes to **Green** (or opposite color)
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4. Numeric Display Check (Accumulator)

- Observe the **Numeric Display** value changing during simulation.
 - Confirm:
 - Address is correct
 - Scaling / Gain / Offset (if used) is working properly
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Stop Simulation

1. Go to **Simulation** → **Stop**.
 2. Exit simulation mode before editing the project.
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Common Simulation Notes (For Students)

- Simulation works **without PLC hardware**.
 - Address mismatch = **no response in simulation**.
 - Always **save the project** before starting simulation.
 - Simulation is only for **testing HMI behavior**, not real PLC outputs.
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